

How the tests are organised and run

Every board owns its tests

How the tests are organised and run in aarch32 /

aarch64

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That is the whole design, and everything below follows from it. There is no shared test directory, no list of test names anywhere in the build, and no capability filter: a board builds the directories it has, and changing a test reaches exactly one board.

1. Where they live

```
micro-os-plus-iii-aarch{32,64}/
├─ src/ include/                the ISA – no board, no test
├─ drivers/ soc/<chip>/          SD, flatfs, DWC2 USB, FatFs; BCM2837 (and RK3506
│                               on aarch32) – formerly micro-os-plus-iii-devices
├─ test/
│   ├─ CMakeLists.txt hw.sh qemu.sh the builder and the two dispatchers
│   ├─ boards/<id>/
│   │   ├─ board.cmake          the board's facts, globbed
│   │   ├─ include/ src/        the board's silicon
│   │   └─ hw.sh qemu.sh        how to run this board's tests
│   └─ <id>/                    ← this board's test applications
│       ├─ <app>/main.cpp        one directory per application
│       ├─ include/ src/        support shared by THIS board's tests
│       └─ tests.cmake          the knobs a directory listing cannot say
```

test/CMakeLists.txt in the port is a loop over test/\${BOARD}/*. It names no test and no board.

Today

port	board	applications
aarch32	<code>rpi-zero-2w</code>	15 — 12 port tests + the harness suites <code>mutex-stress</code> , <code>rtos-apis</code> , <code>cmsis-os-validator</code>
aarch32	<code>rpi3b</code>	15 — the same
aarch32	<code>luckfox-lyra</code>	19 — 11 shared-in-origin, plus this board's own <code>smp_test5</code> ... <code>smp_test7</code> and <code>smp_test_int</code> ... <code>smp_test_int5</code>
aarch64	<code>rpi-zero-2w</code>	15 — as the AArch32 Zero 2 W
aarch64	<code>rpi3b</code>	15 — the same

The Pi boards build every application in two variants, so 30 targets each; the Lyra is hardware-only (\$3), so 19. What each test does is in `tests/TESTS-CATALOG.md`.

Why `rpi3b` has its own `test/` but not its own `src/`

`rpi3b` is the same silicon as `rpi-zero-2w`. Its `board.cmake` points `UOS_BOARD_SRC_DIR` at `../rpi-zero-2w`, so the two boards **share** `src/` and `include/`, compiled with `-DBOARD_RPI3B` — which is exactly how the predecessor's `make BOARD=rpi3b` worked, and that define reaches four places and no others:

file	what <code>BOARD_RPI3B</code> selects
<code>include/led.hpp</code>	a <i>different</i> LED driver — the ACT LED is not a SoC GPIO on this board but VideoCore expander GPIO 130, driven through the ARM↔VC mailbox property channel (tag <code>0x00038041</code>). 91 lines, against 36 for a plain GPIO.
<code>include/uart.hpp</code>	the banner strings
<code>include/smp.hpp</code>	<code>PORT_RAM_END</code> — the window a stack pointer is validated against
<code>src/mmu.cpp</code>	the top of the Normal-memory mapping

plus `linker-rpi3b.ld`, whose RAM region is that same number a third time. **Those last three are one fact written three times and must move together.** They did not, once: `PORT_RAM_END` lived unbranched in a header both boards share, so a Pi 3 B was linked and mapped for 1 GB while every thread stack above 512 MiB looked corrupt to the context switch.

The tests are *not* shared that way, because sharing them is what makes a change to one board able to break another. `_test_dir` is `test/${BOARD}`, never `${UOS_BOARD_SRC_DIR}/test` — which is exactly why `test/rpi3b/` exists beside `test/boards/rpi3b/board.cmake`.

The cost, stated plainly

The twelve port applications exist in up to five copies across the two ports (four Pi boards and, for eleven of them, the Lyra). A fix to a shared test — the `[OS_NCPU] = { 0, 0, 0, 0 }` kind — has to be applied in each copy. That is the trade: duplication bought in exchange for a board that cannot be broken from outside itself. The support code (`test/<board>/include/`, `test/<board>/src/`) is one copy *per board*, not per test, so the duplication is bounded by the number of boards.

2. `tests.cmake` — the three things a directory listing cannot say

Optional. A board whose tests need nothing beyond the defaults has no such file.

```
# Tests that reach the SD card or the USB device controller, and so link
# UOS_BOARD_DEVICES (board.cmake names which driver set that is).
set (BOARD_TEST_NEED_DEVICES
    sd_test smp-mat-sdcard-test smp-num-test smp-pipeline-test usb_test)

# Tests that bring their own start-up code and must NOT also get the board's
# test/<board>/src/ (this is the Lyra's list).
set (BOARD_TEST_SELF_CONTAINED
    smp_test_int smp_test_int2
    smp_test5 smp_test6 smp_test7
    smp_test_int3 smp_test_int4 smp_test_int5)

# Per-test -D flags, carried over from the predecessor's per-test Makefiles.
function (board_test_defines _app _out)
    if (_app STREQUAL "usb_test")
        set (${_out} LED_BLINKS=3 LED_ON_MS=40 LED_OFF_MS=40
            LED_GAP_MS=300 USB_FORCE_FS PARENT_SCOPE)
    endif ()
endfunction ()
```

`BOARD_TEST_NO_KERNEL` names the tests that compile **no kernel at all** — the bare-metal probes that run before a scheduler exists. They link the port's bare target (`UOS_PORT_BARE_LIB`) instead of its full one and are declared `NO_KERNEL`, so no kernel, no port scheduler and no `OS_NCPU` reach them. Only the Cortex-M port has such tests today; a port that declares no bare target rejects the request rather than quietly linking a kernel.

`BOARD_TEST_SELF_CONTAINED` exists because a test carried over whole from the predecessor defines its own `smp_install_boot_threads()`. Linking the board's shared copy as well is a duplicate definition, and picking one of them would mean editing a test that already passed on hardware.

Three more hooks answer "what if **one** test disagrees with its board", and each defaults to the board's own answer, so a board that does not use them never sees them:

```
# A linker script instead of the board's -- the RP2350B PSRAM tests link
# .data/.bss/heap into external memory at 0x11000000.
function (board_test_linker _app _out) ... endfunction ()

# How many CPUs ONE test runs on. uos_add_app() derives
# OS_USE_SMP_SCHEDULER from NCPU, so returning 1 on a two-core board is how a
# single-core test is expressed.
function (board_test_ncpu _app _out) ... endfunction ()

# Compile options on the test's OWN sources, not on the kernel it links --
# the RP2350 nested-interrupt tests need "-g0 -mlong-calls" because their
# handlers live in .data.
function (board_test_options _app _out) ... endfunction ()
```

`board_test_linker()` is also the per-test linker override that `smp-pro-cons-test` wanted here: it carries its own script in the predecessor, and until now the build had no way to say so.

Which tests a board has is not configured — it is observed

There is no `if (BOARD STREQUAL ...)` and no capability list gating tests. A board with no SD card has no `sd_test` directory. A single-CPU board will have no `smp_*` directories. This is what makes the `cortexm` port tractable, with six boards, half of them non-SMP.

3. The two variants

Every application is built once per variant, from the same sources:

variant	define	linker	for
<code><app>-qemu</code>	<code>QEMU_BUILD</code>	<code>UOS_BOARD_LINKER_QEMU</code>	the emulated suite
<code><app>-hwd</code>	<code>HW_BUILD</code>	<code>UOS_BOARD_LINKER_HW</code>	real silicon, over OpenOCD

This file describes the AArch32 and AArch64 ports. The Cortex-M port carries the same `test/CMakeLists.txt`, the same dispatchers and the same hooks — `board_test_defines()`, `board_test_sources()`, `board_test_includes()`, `board_test_linker()`, `board_test_ncpu()`, `board_test_options()`, `BOARD_TEST_SELF_CONTAINED` — so everything here about layout and per-test composition applies to it too. The last three hooks arrived *from* that port, where one board's tests disagree with their board about CPU count, memory map and compile options. Two of its six boards, `pico2` and `pico2-rp2350b-psram`, also build `-qemu` images for the few tests QEMU's generic Cortex-M can run; the other four are hardware-only. See [cortexm-port.md](#).

The POSIX port carries the same loop and the same six hooks with one difference, and it is a difference in the machine rather than in the method: there is **one** variant, `host`, because the host *is* the machine — there is nothing to emulate and nothing to flash, and a test is simply an executable. It is deliberately not a `HW_BUILD`: that define means "use the board's real FAT32 boot partition", while the SD tests' other branch is flatfs over a plain image file, which is exactly what a host-file back-end is. Its dispatcher is `test/run.sh` rather than `qemu.sh` / `hw.sh`. See `posix-arch-port.md`.

How many variants is the board's decision, not this file's. A board that sets `UOS_BOARD_LINKER_QEMU` gets both; a board that leaves it unset gets `hwd` only, because there is nowhere for the other image to run. The absence *is* the declaration — there is no separate flag that could disagree with it.

The Luckfox Lyra is that case, and deliberately. QEMU has no RK3506 machine, and the generic `virt` models none of the blocks that make the board worth having — the DWC2 gadget, the DesignWare SD host, the SRAM mailbox, the Cortex-M0. An emulated Lyra would be a generic Cortex-A7 wearing the name. So the Lyra builds 19 images, not 38, and `test/qemu.sh` points at `hw.sh` when asked for it. The Pi boards build both variants and run the emulated suite.

`HW_BUILD` makes the SD tests use the board's existing FAT32 boot partition. `UOS_DEBUG_BOOT=ON` additionally compiles the early-boot assembly markers; it is off by default because each marker is a semihosting trap and under a JTAG probe each one costs real time.

4. Running them

Every board has the same two scripts, with the same interface. The port-level `test/hw.sh` and `test/qemu.sh` are dispatchers that contain no board names: `BOARD` picks the directory.

```
# emulated
BOARD=rpi3b test/qemu.sh           # the whole suite
BOARD=rpi3b test/qemu.sh smp_test0 # one test, output live

# hardware
BOARD=rpi3b test/hw.sh list        # what this build has
BOARD=rpi3b test/hw.sh smp_test0   # run one
BOARD=rpi3b test/hw.sh smp_test0 300 # ...with a 300 s budget

# the POSIX host (micro-os-plus-iii-posix-arch)
BOARD=native test/run.sh           # the whole suite
BOARD=native test/run.sh smp_test2 # one test, output live
```

or call the board directly — `test/boards/rpi3b/hw.sh smp_test0`. An unknown board gets the available ones listed rather than a wrong branch taken.

Each board's script holds only that board's facts and delegates the session itself to one shared runner in the kernel repository:

<code>test_smp1/run-qemu.sh</code>	runs every <code>*-qemu.bin</code> in a build directory, reports <code>PASS</code> / <code>SKIP</code> / <code>FAIL</code> from the <code>RESULT:</code> line each test prints, and seeds an SD image for the tests that need one
<code>test_smp1/run-host.sh</code>	the same, for the POSIX port: runs every <code>*-host</code> executable, same timeout table, same verdicts, same summary line — <code>run-qemu.sh</code> with the emulator taken out
<code>test_smp1/run-hw.sh</code>	halts the cores, enables semihosting, <code>load_image</code> , resumes

Those three files are the same for every board of every port. Nothing board-specific is in them and nothing generic is in the board scripts.

What a board script declares

```

UOS_HW_ENTRY=0x1003c          # where the image is linked
UOS_HW_SPIN_WORDS=0           # __smp_spin words to zero before release
UOS_HW_RESUME=cpsr-first      # cpsr | pc | entry | cpsr-first | pc-first
UOS_HW_NCPU=4
UOS_HW_TARGET_FMT="bcm2837.cpu%d"
UOS_HW_CFG=.../openocd-jlink-rpi3.cfg
UOS_HW_CFG_INIT=1             # 1 when the config is purely declarative
UOS_HW_PRELOAD="...Tcl..."  # run between halt and load (the Lyra's
                                # MMU/cache sanitize)

```

The AArch64 Pi differs from the AArch32 Pi on the same silicon by two of these: entry `0x80000`, and `pc-first` (core 0 is started by PC, because a core keeps the exception level it was halted in). Both Pi ports' OpenOCD configs are purely declarative (`CFG_INIT=1`); the AArch32 ones used to run `init` and `halt` themselves, and that version is kept as `openocd-jlink-rpi3.cfg.bak` / `openocd-olimex.cfg.bak`.

On both Pi ports only core 0 is started (`cpsr-first` / `pc-first`); cores 1–3 are resumed where they were halted — after the watchdog reset, in the firmware's own loop — and the kernel releases them itself once it has written their `__smp_spin` slot, so no spin words are zeroed (`UOS_HW_SPIN_WORDS=0`). That needs an SD card whose kernel leaves cores 1–3 in that loop and boots the port's width (`arm_64bit=0` for AArch32, `arm_64bit=1` for AArch64). The previous flow — all four cores started at the entry, after zeroing 4 (AArch32) or 8 (AArch64) `__smp_spin` words — is kept in each port's `test/boards/rpi-zero-2w/hw.sh.bak`.

5. Two rules the runners keep

Pure OpenOCD. No GDB anywhere — the configs set `gdb port disabled` outright. OpenOCD halts, loads and resumes; the application talks over the UART and semihosting.

No redirection. OpenOCD's output and the board's semihosting go to your terminal as they happen; the copy teed to `.hw-logs/<app>.log` exists only so the loop can match the verdict. The UART is never opened by any script, so `tio -b 115200 /dev/ttyACM0` in your own window is never fought over. A QEMU suite of a dozen tests is read from its summary, so those stay captured — but `qemu.sh <app>` streams, because that is the run you are watching.

One test per power cycle, on hardware. Every run is a `load_image` into RAM over whatever the previous test left there, and neither board has a reset a script can drive: the Pi has no SRST and its Cortex-A53 debug target has no reset method, and the Lyra's secondaries are released once, by clearing `__smp_spin`, which a second load cannot undo. `run-hw.sh` refuses a suite for this reason and takes exactly one test.

6. Adding a test, adding a board

A test, for one board: make `test/<board>/<name>/` with a `main.cpp`. That is all — it is globbed. If it needs the SD card, add its name to `BOARD_TEST_NEED_DEVICES`; if it brings its own start-up code, to `BOARD_TEST_SELF_CONTAINED`.

A test, for several boards: copy it into each board's `test/`. There is deliberately no mechanism to avoid this.

A board: make `test/boards/<id>/` with `board.cmake`, `include/`, `src/`, `hw.sh`, `qemu.sh`, a linker script and an OpenOCD config, and `test/<id>/` with its applications. Nothing outside those two directories is edited — not `CMakeLists.txt`, not `test/hw.sh`, not another board. `board.cmake` is globbed, and the configure step names any fact it forgot; `src/board-contract.cpp` compiles nothing and fails the build for a board that left `PORT_RAM_BASE`, `PORT_RAM_END`, `OS_NCPU`, `PORT_GREETING`, the banner strings, `SEMIHOST_TRAP_CHOSEN` or `OS_SMP_IPI_SGI` undeclared.

A board sharing another's silicon points `UOS_BOARD_SRC_DIR` at it and `#ifdef`s the differences, as `rpi3b` does — but it still gets its own `test/`.

7. Known gaps

- `smp-pro-cons-test` carries its **own** linker script in the predecessor (`smp-pro-cons-test/linker{,-rpi3b}.ld`). The per-test override now exists (`board_test_linker()`, §2), but no board uses it for this test: it links the board's script.

- The six predecessor Lyra tests once parked in `test/luckfox-lyra/.pending/` now build and are ordinary test directories (`smp_test5` ... `smp_test7`, `smp_test_int3` ... `smp_test_int5`); the `.pending/` directory is gone.
- `smp_test4` has not been made to pass on the Lyra. The shared `smp_test4` is the Pi's no-affinity load-balancing test; the predecessor's Lyra `smp_test4` is a different program entirely, an 865-line CNTPCT benchmark.